B2.2 Organelles and compartmentalization

NAME	DATE	TOTAL
		_____ / 24

About 25 minutes, or one section at a time. The number of marks shows how many separate points to make.

Section A: Recall

5 MARKS

1 **DEFINE** organelle. [B2.2.1] [1]

2 **STATE** two cell structures that are not organelles. [B2.2.1] [2]

3 **NAME** one organelle with two membranes and one with no membrane. [B2.2.1] [2]

Section B: Understanding

9 MARKS

4 **EXPLAIN** the advantage of separating the nucleus and cytoplasm into compartments. [B2.2.2] [3]

5 **OUTLINE** two advantages of compartments in the cytoplasm. [B2.2.3] [2]

6 **EXPLAIN** why digestive enzymes are kept inside lysosomes. [B2.2.3] [2]

7 **OUTLINE** how a phagocytic vacuole forms and what happens inside it. [B2.2.3] [2]

Section C: Working with data

6 MARKS

All data in this section are invented for practice. The patterns are realistic.

8 [B2.2.3]

The pH inside a food vacuole of a unicellular organism was measured over time.

TIME AFTER FOOD ENTERS / MIN	0	5	10	20	40
PH INSIDE THE VACUOLE	7.0	4.5	3.0	6.5	7.5

a **DESCRIBE** the change in pH. [2]

b **SUGGEST** why this change would be damaging if it took place throughout the cytoplasm. [2]

c **SUGGEST** why different enzymes might act at different times in the vacuole. [2]

Section D: Longer answer

4 MARKS

9 **EXPLAIN** why eukaryotic cells with compartments can be more efficient than cells without them. [4]

[B2.2.2, B2.2.3]